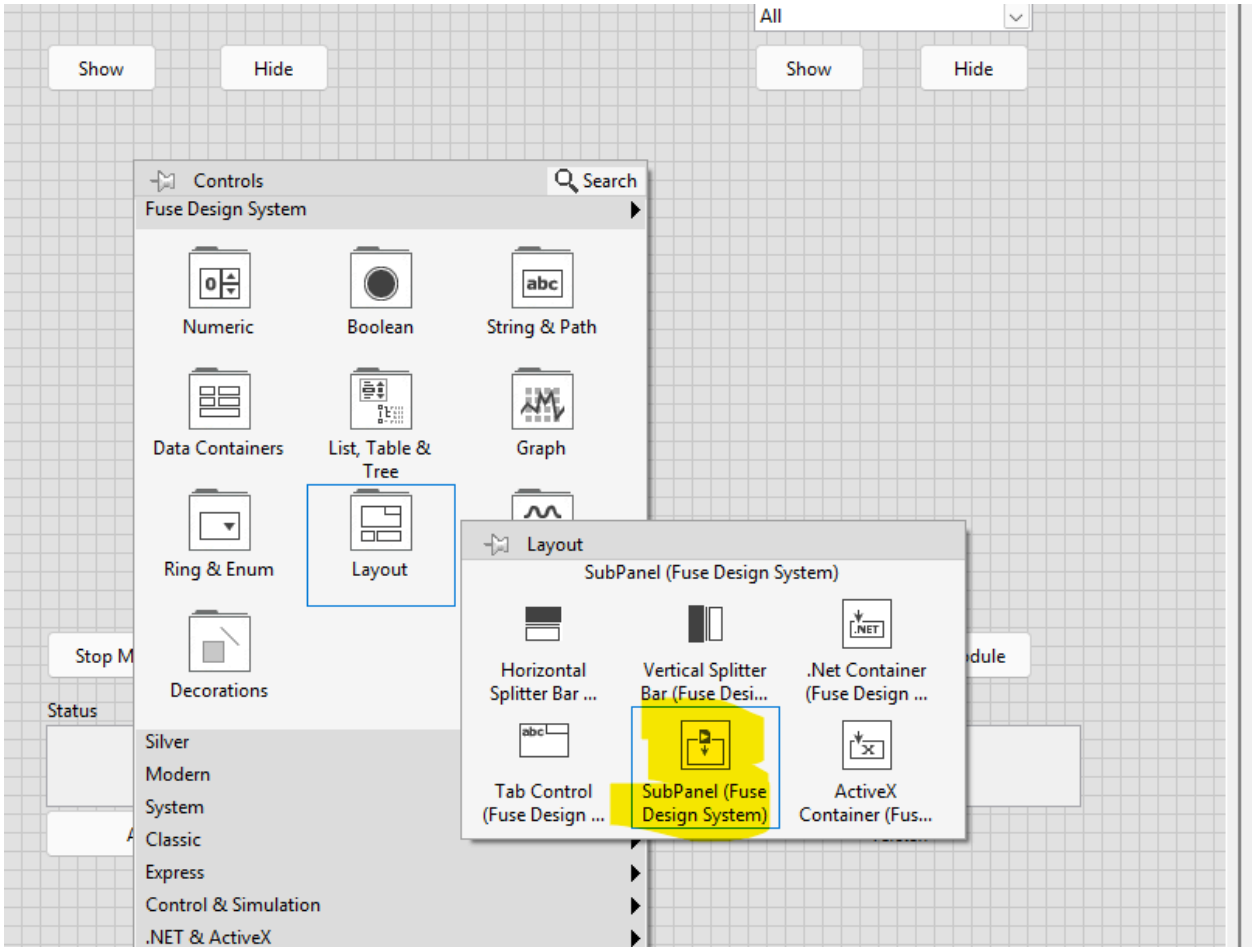
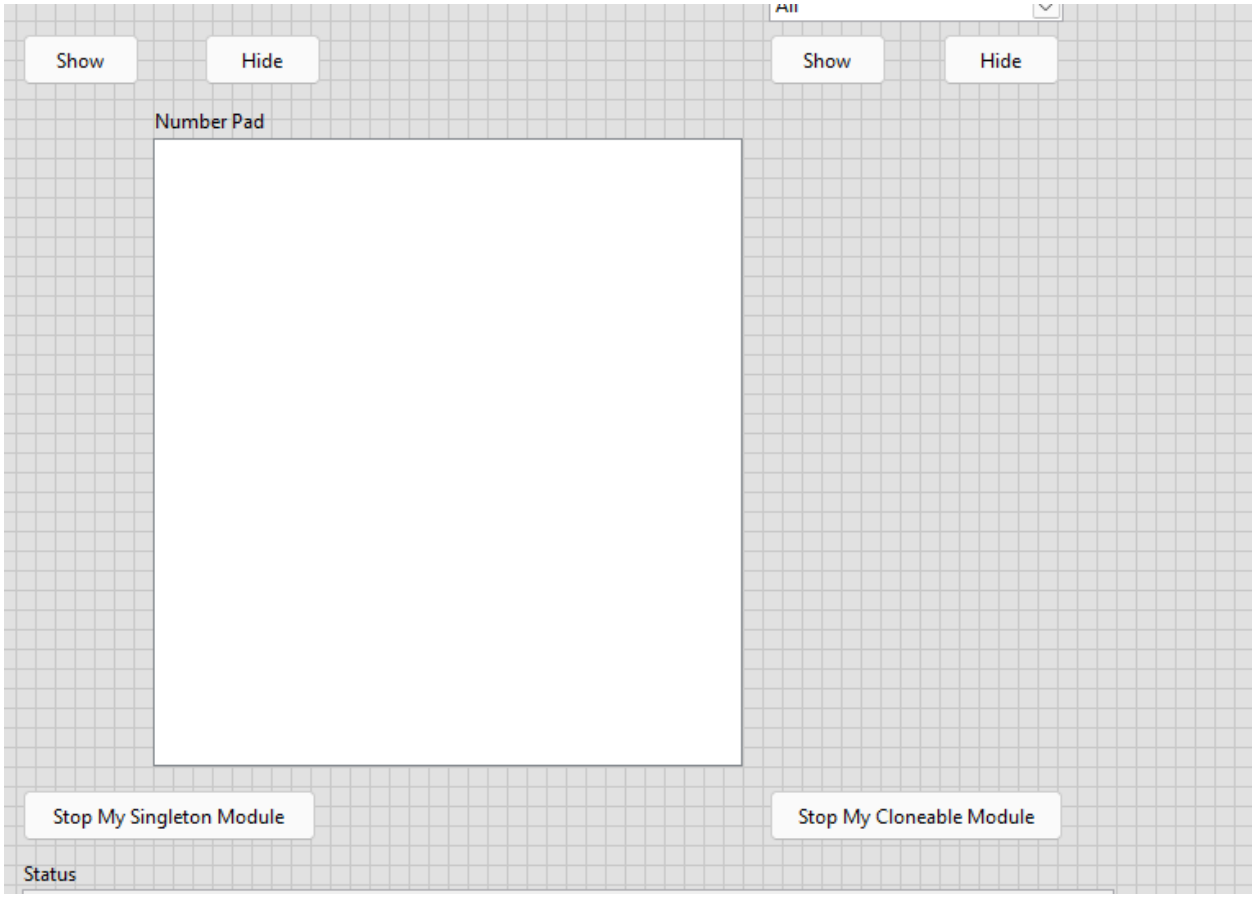
Method 1

Method 1: Requesting a VI Reference from the VI (module) you wish to insert into a subpanel. Method 2 is in a different tab to the left.

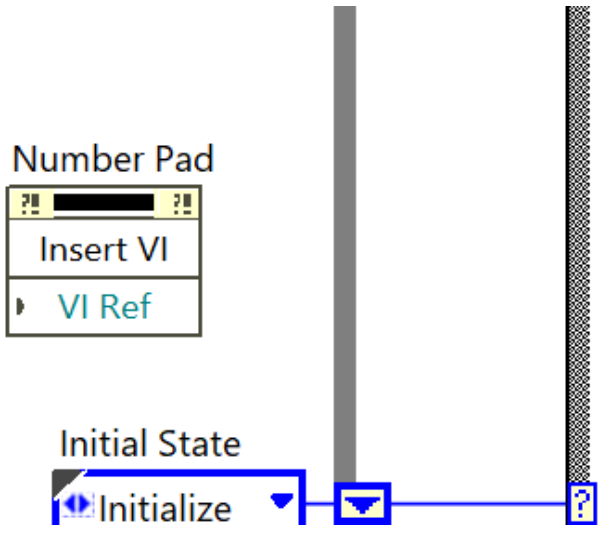
You can **zoom** into this document to see the details better.



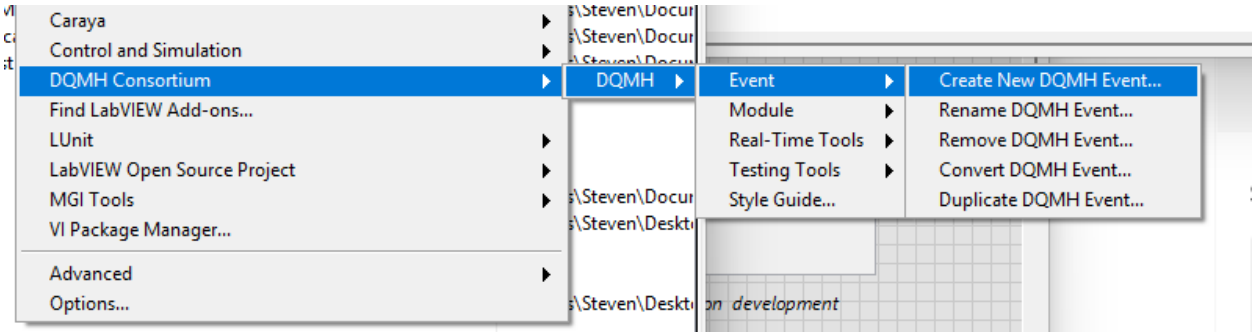
1. Start by placing a SubPanel on the front panel.



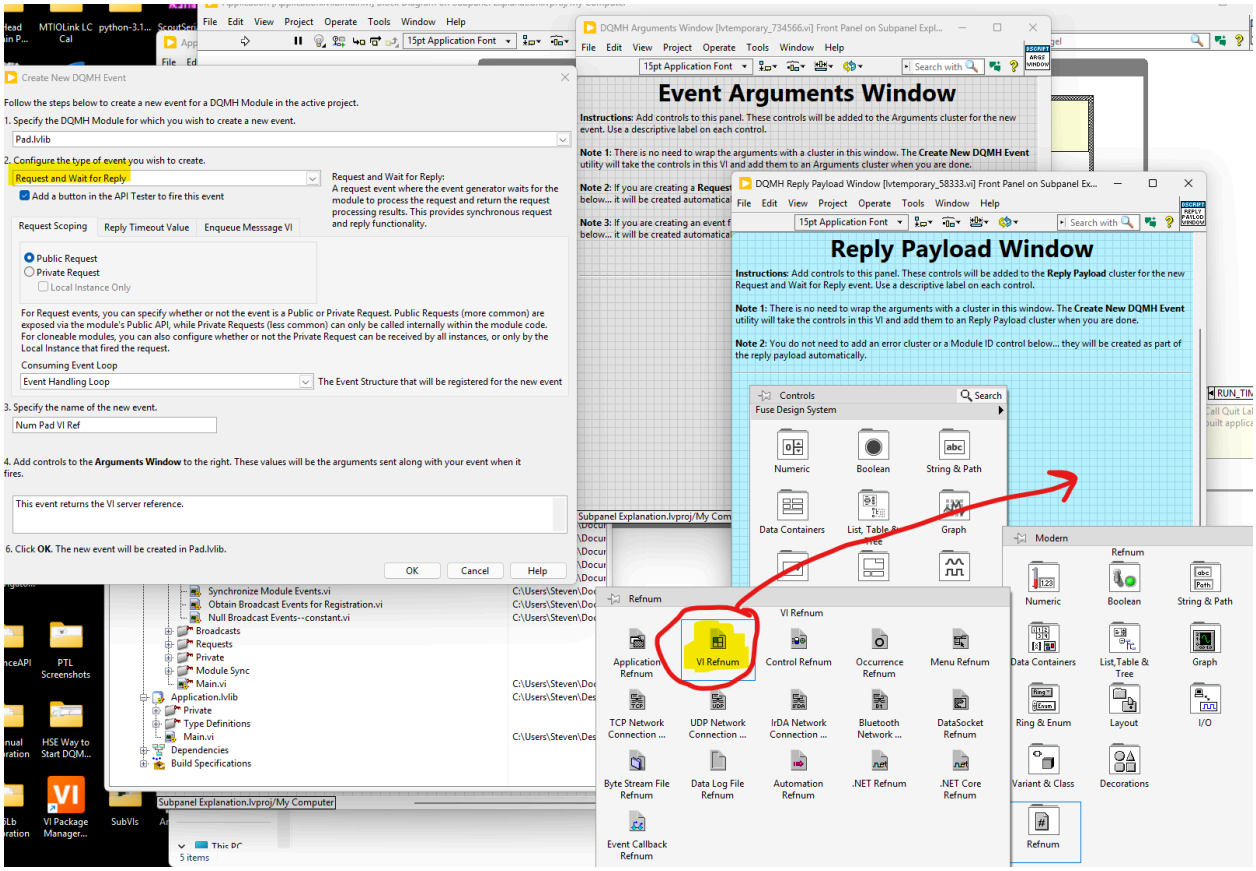
2. Don’t forget to give it a descriptive name. It can also be hidden like all labels. The descriptive name is useful for the reference nodes.



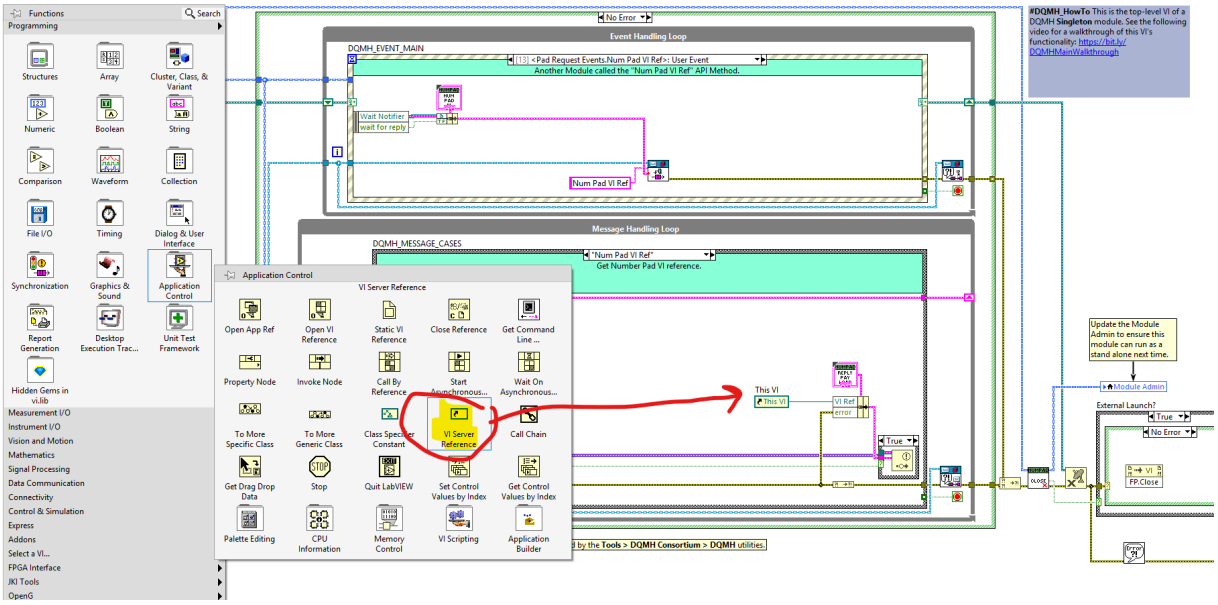
3. When you make a SubPanel, it will give you one of these Invoke Nodes. Delete it.



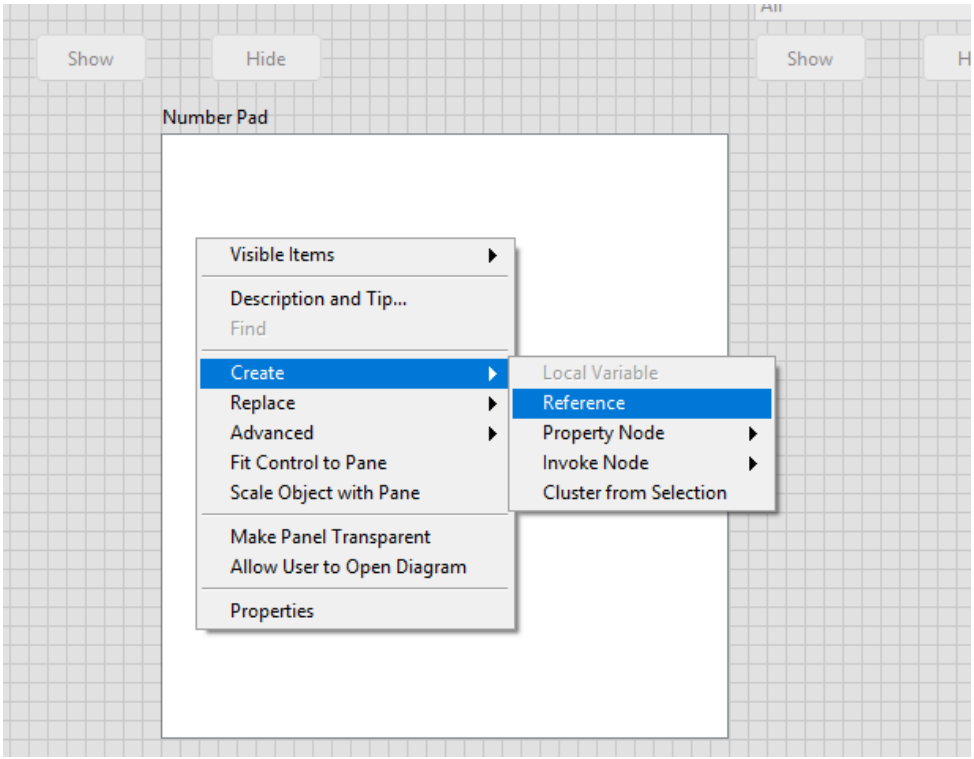
4. Create a new Request and Wait for Reply event **for the module that you will be inserting into the subpanel**.



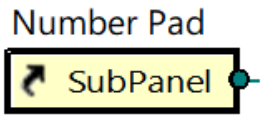
5. In the Reply Payload Window, place a VI Refnum.



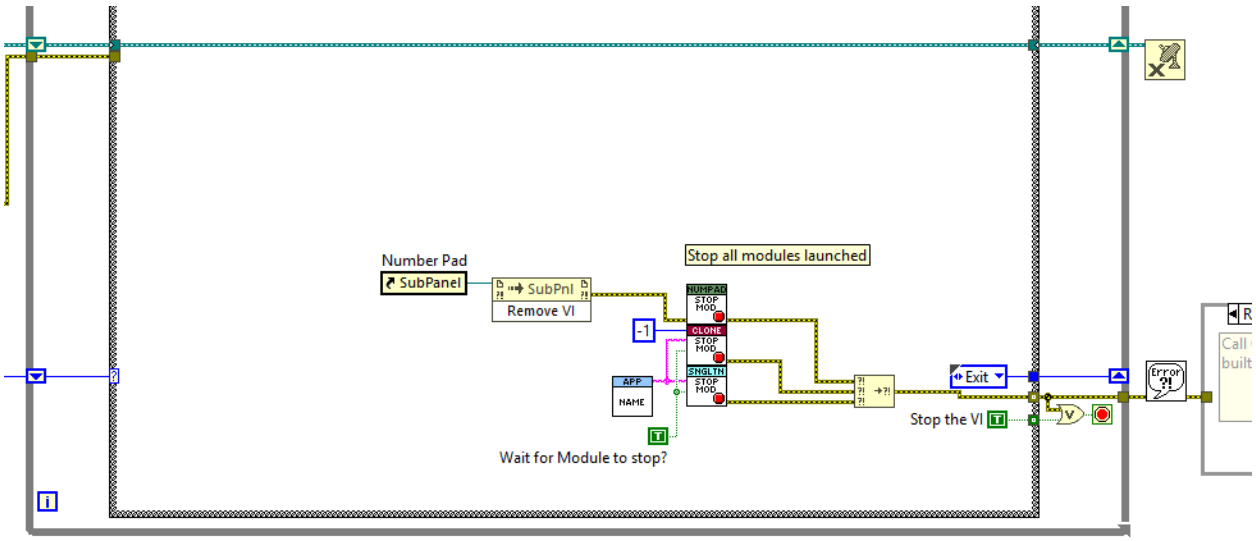
6. Return a VI Server Ref when this event is triggered.



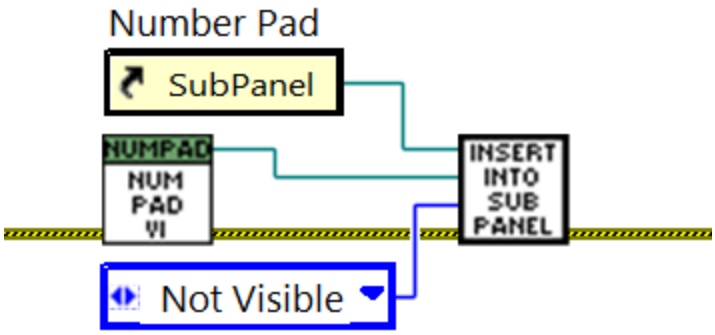
7. On the calling VI, create a reference to your SubPanel.



8. You'll end up needing at least two of these references.

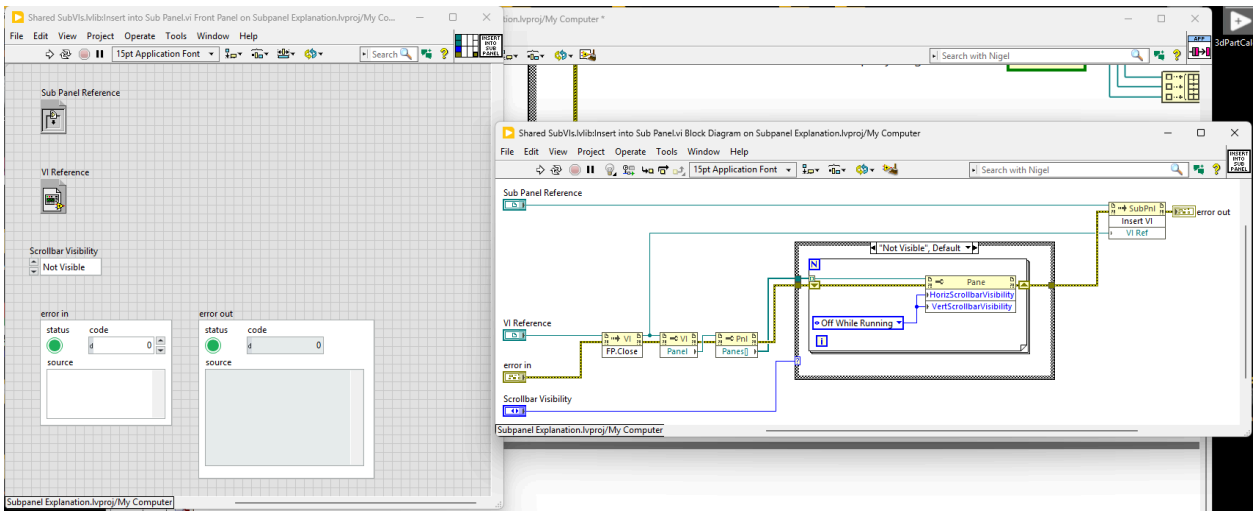


9. Let's "start by stopping". Wherever your calling VI is stopping, make sure you remove your module from the SubPanel before shutting the module down with an Invoke Node. *Hot tip if you have multiple subpanels, build an array of the SubPanel refs and wire them to a For Loop with the Remove VI invoke node in it.

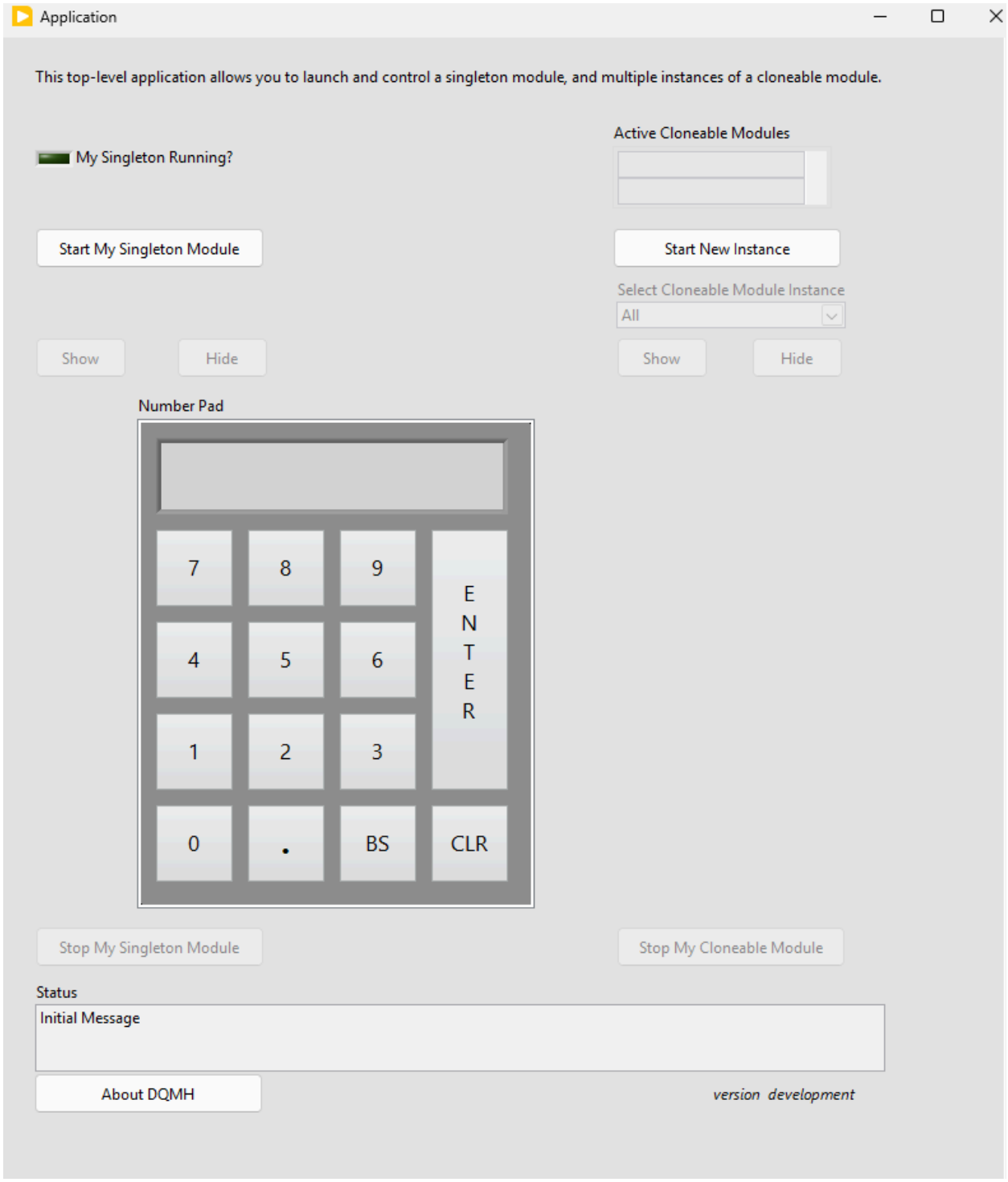


10. Here's the fun part. After starting your module, use the Request and Wait for Reply event you created earlier to return the VI Server Ref for the module you want to insert into a SubPanel. You'll also need the reference to your SubPanel. Don't worry, I'll show you what is in this subVI.

Where and when do you do this? That's up to you. You can link this to an event, put it in your initialization case, or however else you want to go about inserting your module into a subpanel.



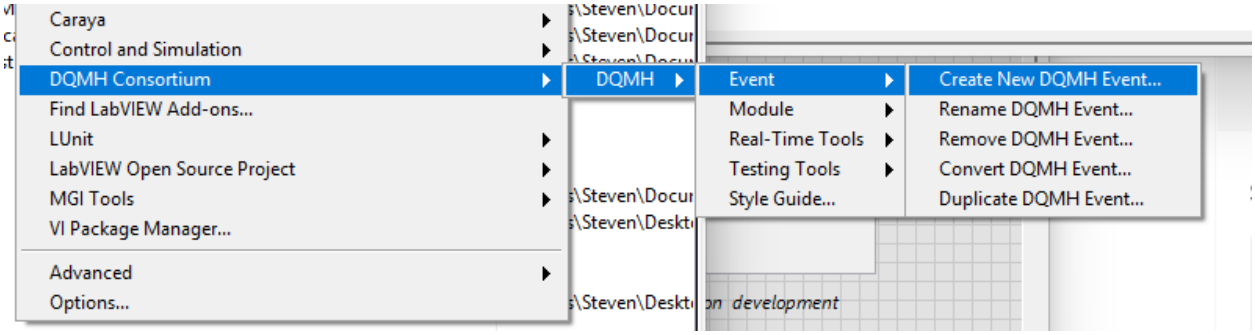
11. Here's what is going on in the Insert Into Sub Panel.vi. I would recommend doing something similar. It is very useful to have it as a subVI.



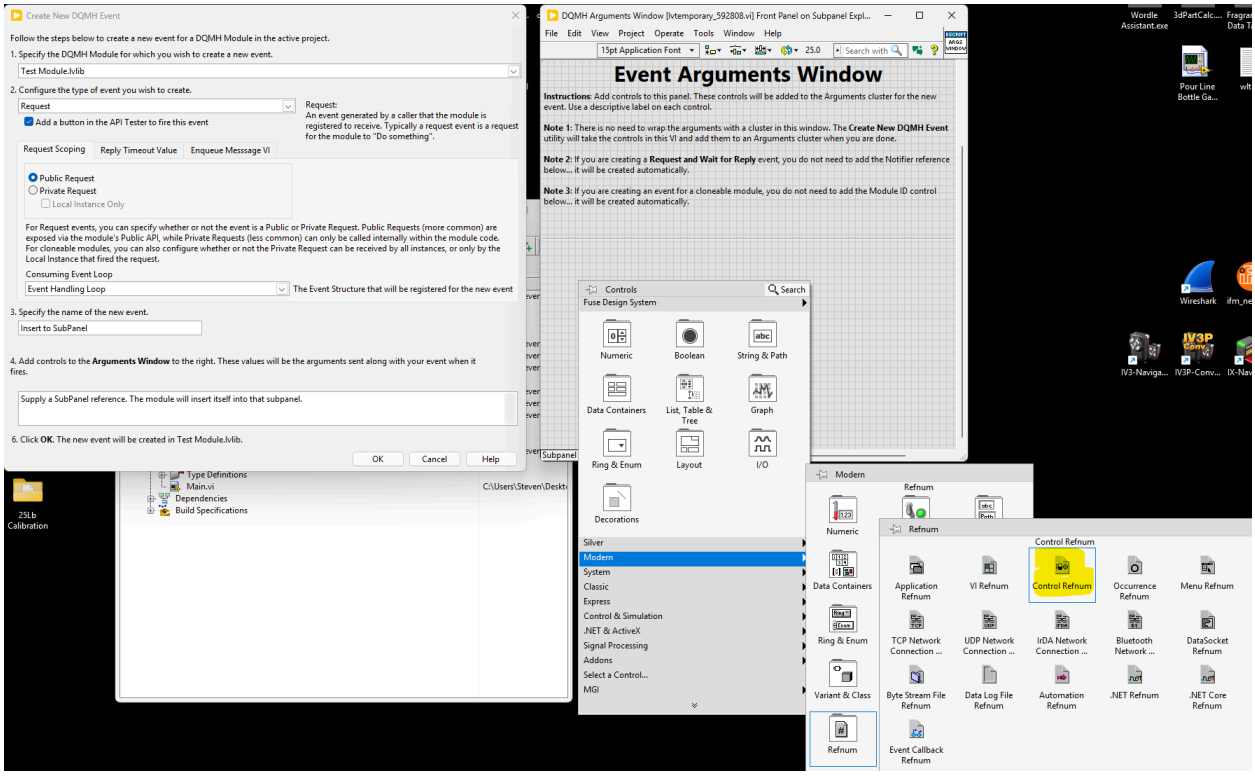
12. And now you'll have a DQMH module in a SubPanel of your calling VI.

Method 2

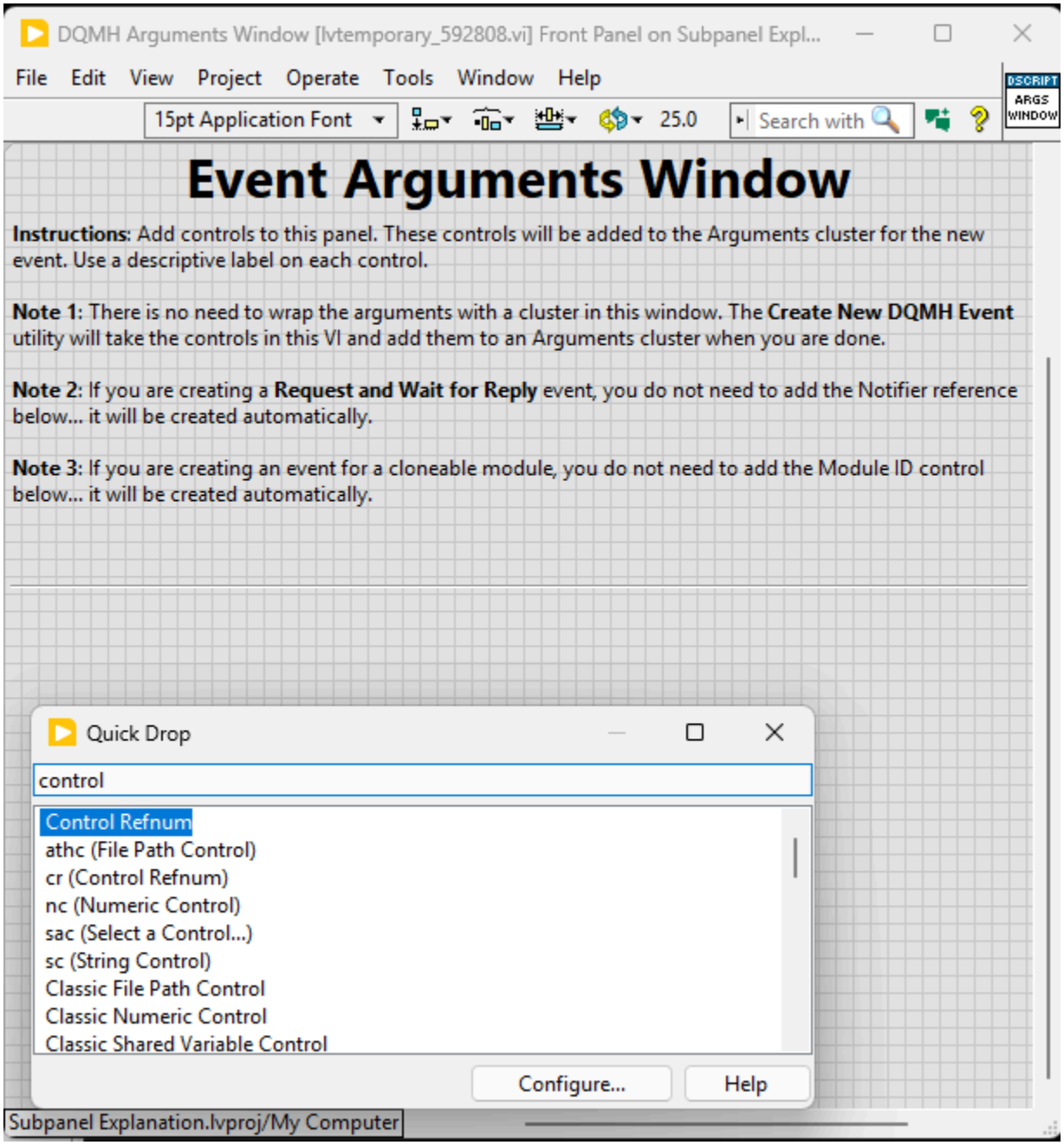
Method 2: Sending a SubPanel Reference to the module you wish to insert into a subpanel via a Request event. The module receiving the subpanel reference will handle insertion and removal of itself.



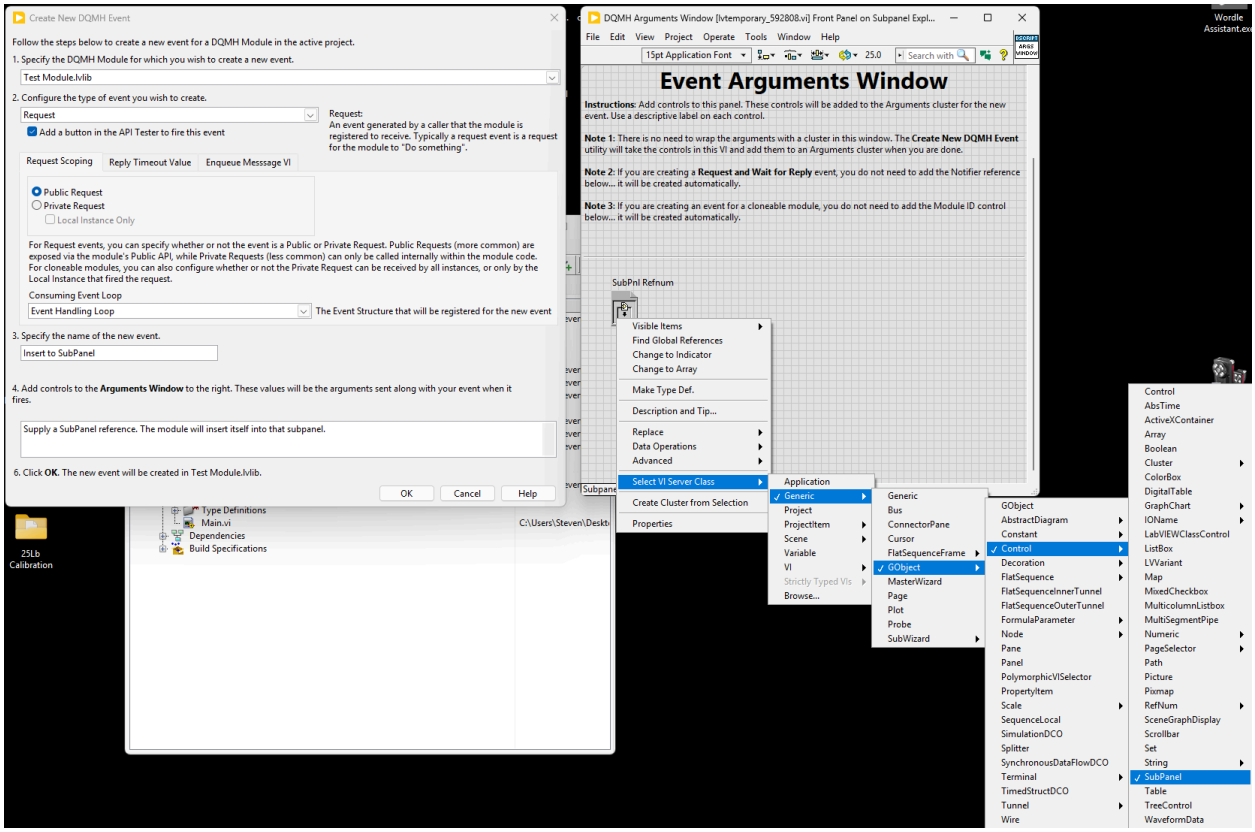
1. Create a new DQMH Event **<u>for the module you want inserted into a subpanel.</u>**



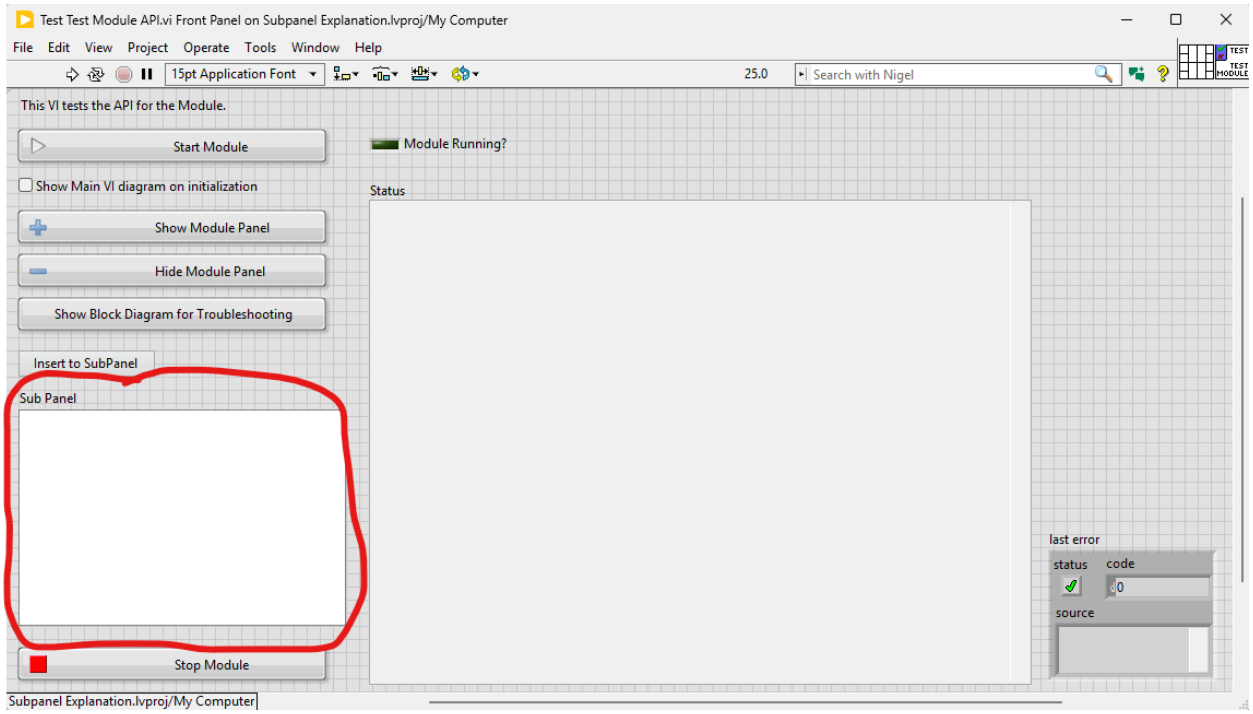
2. Place a “Control Refnum” in the Arguments Window.



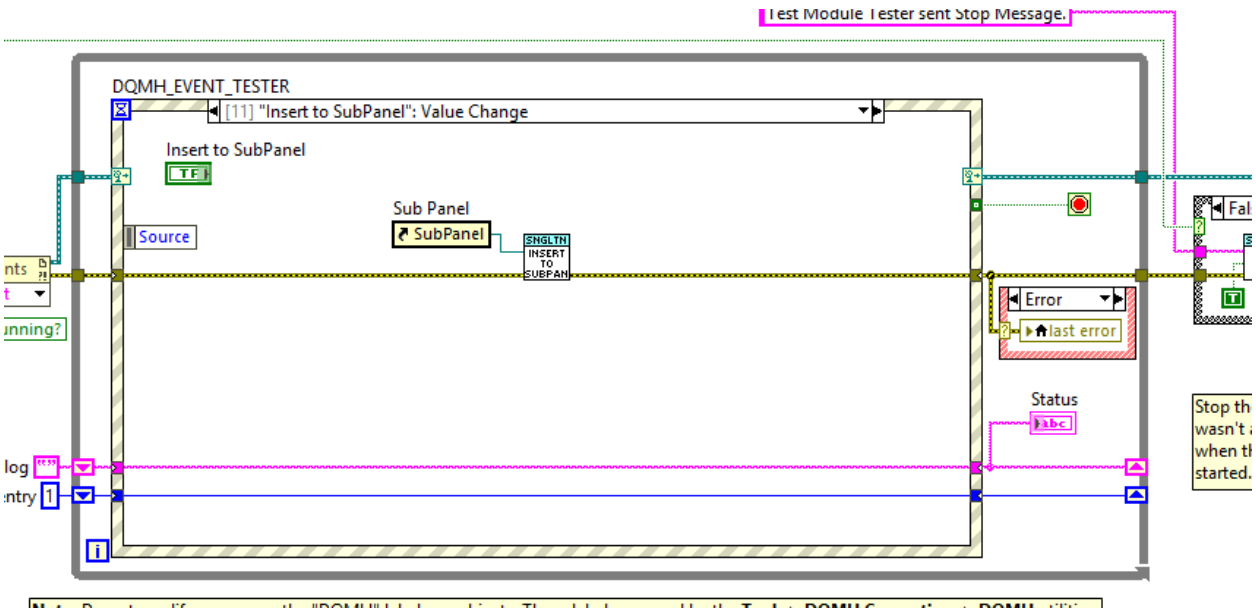
2a. Or, alternatively, just use Quick Drop.



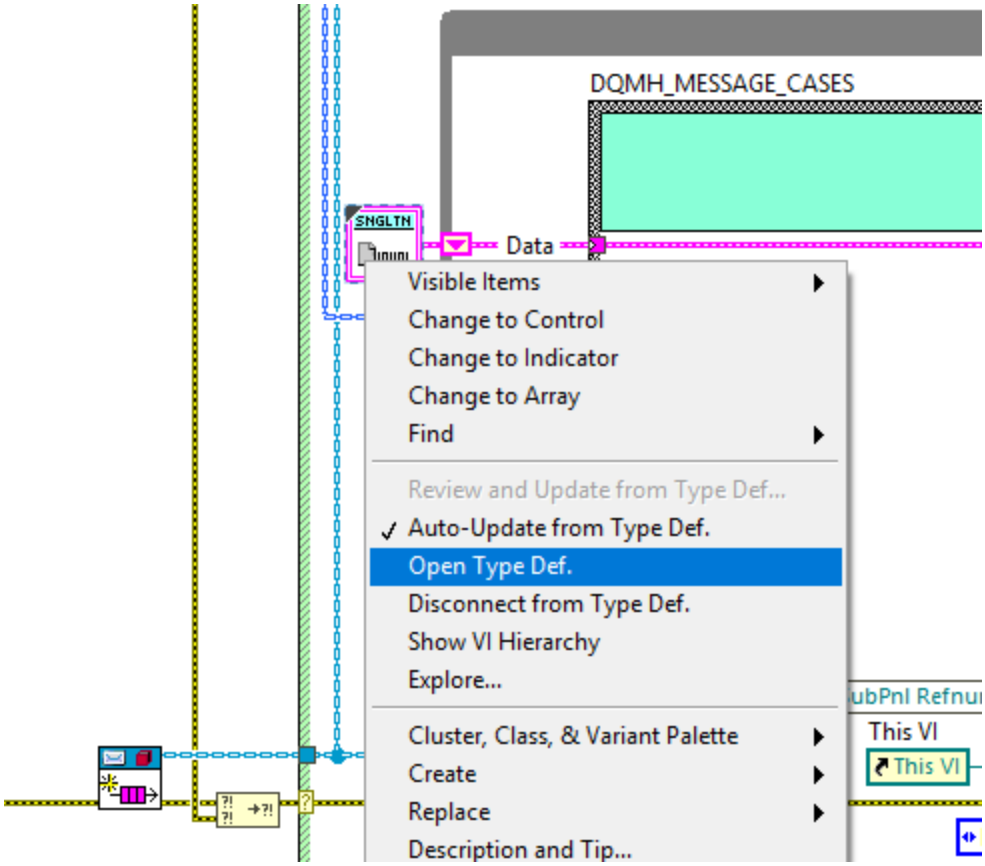
3. Make that Control Refnum a “SubPnl Refnum”. Make sure the event type is “Request” and Public. Give it a descriptive name and good documentation. Then click “OK” to create the event.



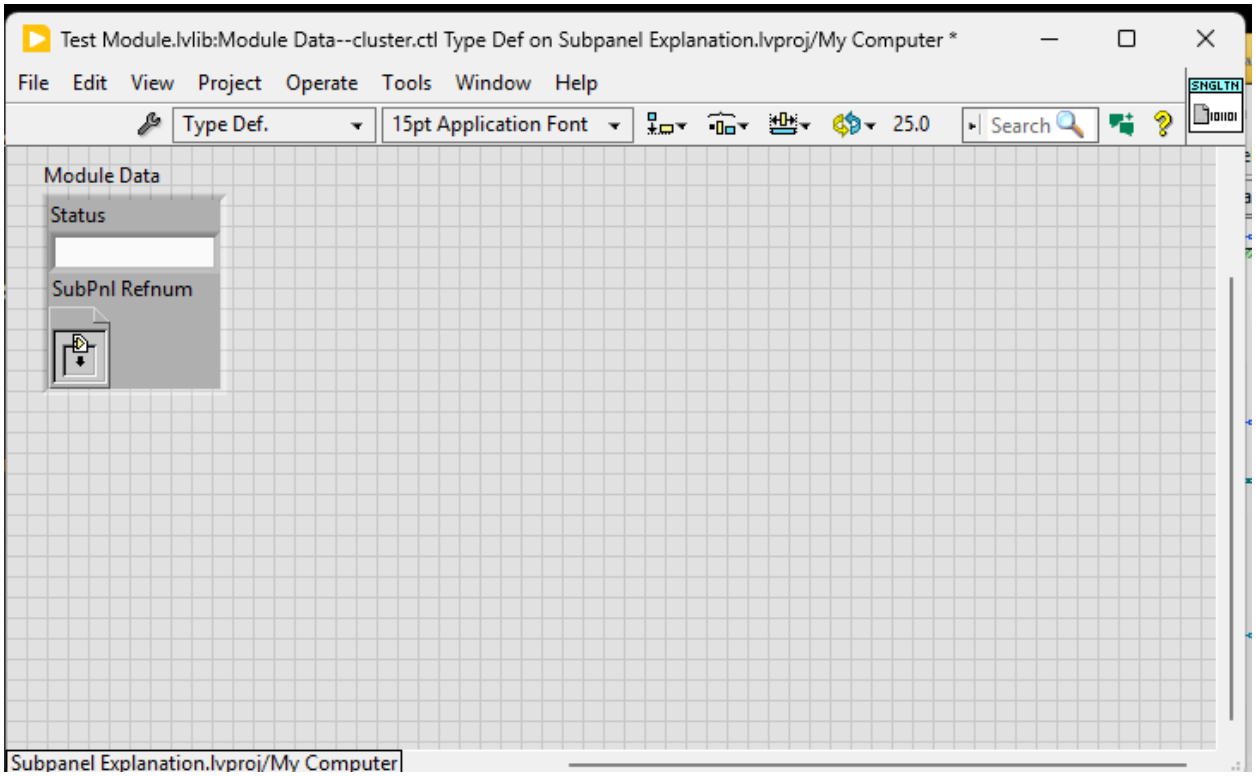
4. I'll use the tester for this explanation. Place a Sub Panel on the front panel of the tester.



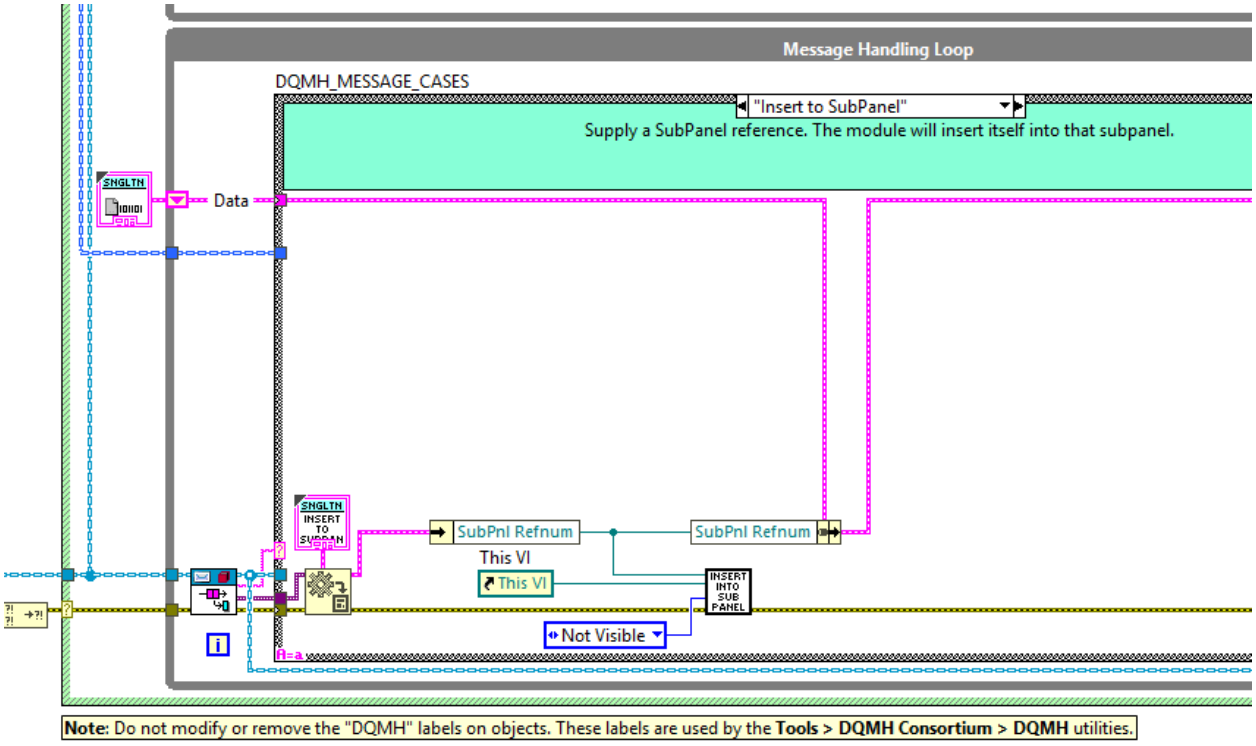
5. Create a reference for that subpanel and wire it into the request event on the block diagram of the tester.



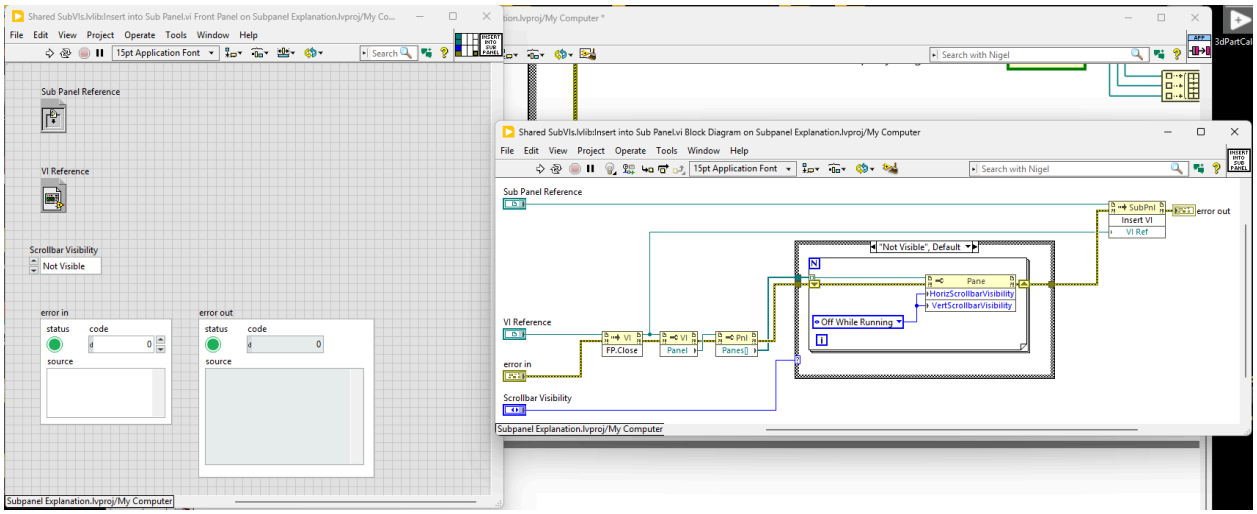
6. On your module's Main.vi Block Diagram, open the Data Cluster Type Def.



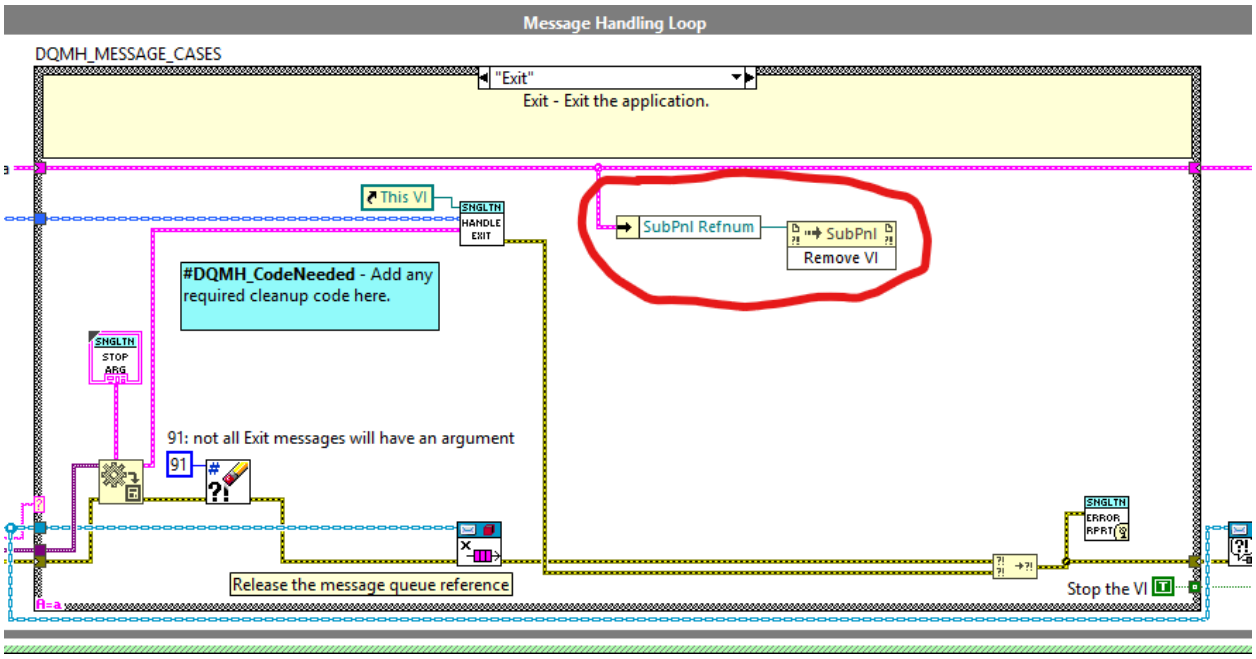
7. Insert a “SubPnl Refnum” into the Module Data cluster like you did in steps 2 & 3. Apply changes and Save the Type Def.



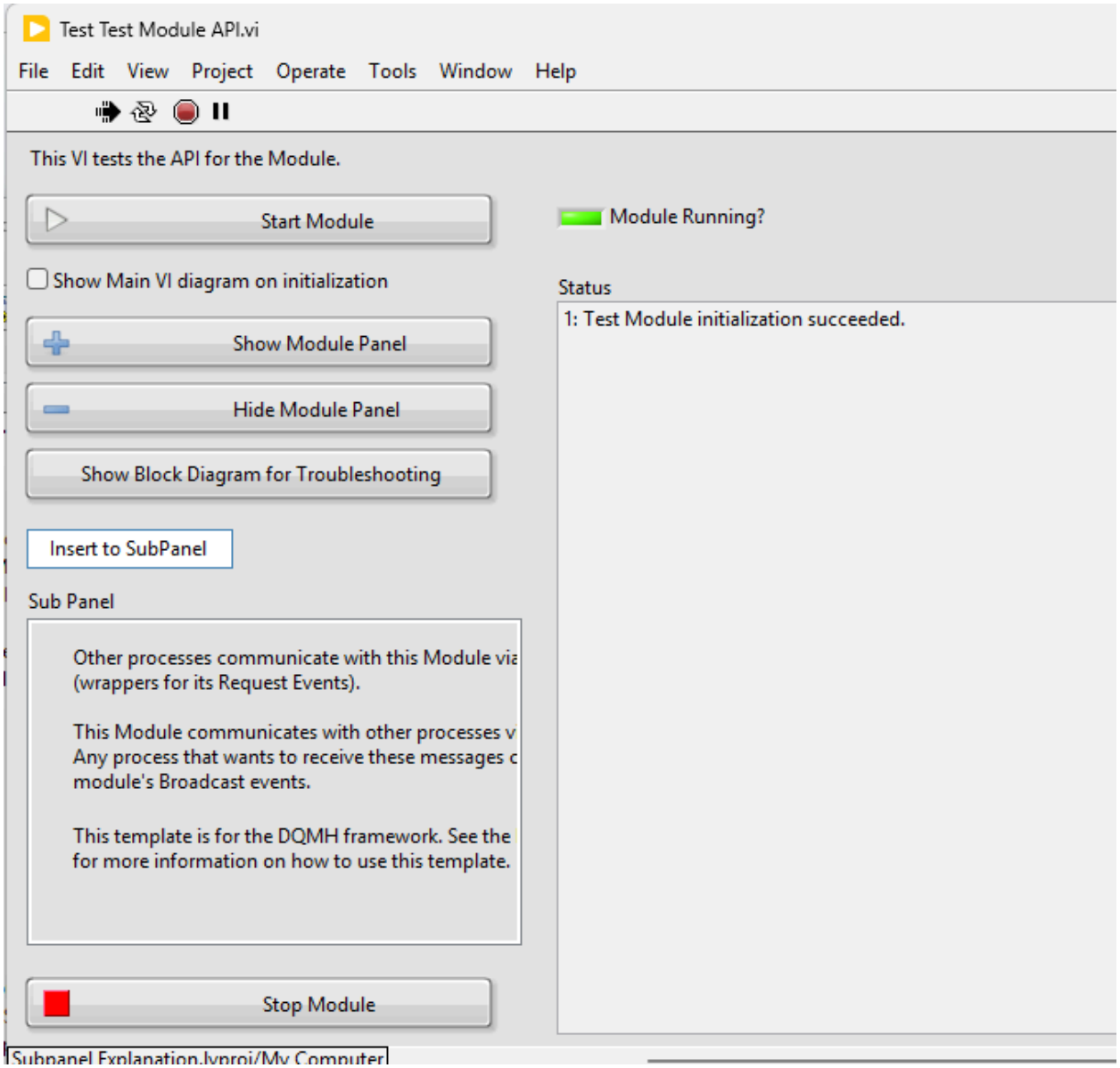
8. Take the subpanel refnum that is supplied via the request event and use it to insert the module into the subpanel. Make sure you bundle that reference to the module's data cluster so that you can remove the module from the subpanel on exit.



*Here's what's going on in the "Insert Into Sub Panel.vi". I would recommend doing something similar.



9. In the Exit case, make sure to unbundle the subpanel ref and wire it to an invoke node that removes the VI from the subpanel before exiting.



10. Test this new Request event in your tester. You should see the Sub Panel populate with your DQMH module when you fire the event. When you click “Stop Module”, the DQMH module should remove itself from the Sub Panel. If the entire Tester FP closes, you didn’t properly remove the DQMH module from the Sub Panel first.